

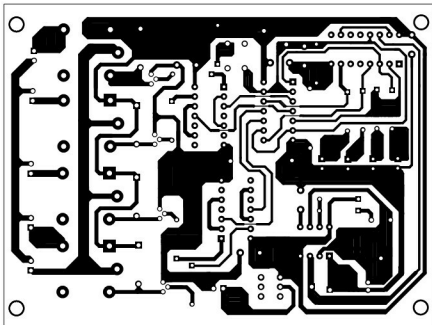


Fig. 2: Actual-size, single-side PCB for one-switch switchboard

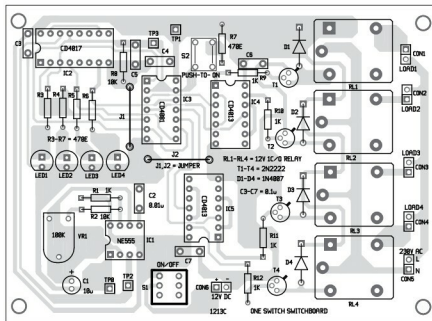


Fig. 3: Component layout of the PCB

nected to first inputs of AND gates N1 through N4 of IC3. So the first inputs of the AND gates go high in a sequence as the counter advances. The second input of each AND gate is tied together to switch S2.

Now if you momentarily press switch S2, the second input of each gate goes high and the output of that AND gate goes high whose first input is held high by IC2. The outputs of

these AND gates are fed to four different D flip-flops in IC4 and IC5. The flip-flop which receives a high clock pulse changes its state. The outputs of D flip-flops control four relays (RL1 through RL4) using transistors T1 through T4. Freewheeling diodes D1 through D4 protect the transistors from back EMF.

Working of the circuit is simple. When LED1 glows and you press

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- IC2 - CD4017 decade counter
- IC3 - CD4081 quad AND gate
- IC4, IC5 - CD4013 dual D flip-flop
- T1-T4 - 2N2222 npn transistor
- LED1-LED4 - 5mm red LED
- D1-D4 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2, R8 - 10-kilo-ohm
- R3-R7 - 470-ohm
- R9-R12 - 1-kilo-ohm
- VR1 - 100-kilo-ohm preset

Capacitors:

- C1 - 10µF, 25V electrolytic
- C2 - 0.01µF ceramic disk
- C3-C7 - 0.1µF ceramic disk

Miscellaneous:

- S1 - On/off switch
- S2 - Push-to-on switch
- CON1-CON6 - 2-pin connector
- RL1-RL4 - 12V, 1C/O relay
- 12V battery/adaptor

switch S2 momentarily, LOAD1 is switched on. Similarly, if you want to switch on the other load you wait for the corresponding LED to glow and then press switch S2. To switch on or off all the loads together, keep switch S2 pressed till LED4 glows.

Construction and testing

An actual-size, single-side PCB for one-switch switchboard is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic case. Fit all the four LEDs on the front panel and label the corresponding loads below each LED. Fit on/off switch S1 and push-to-on switch S2 also on front panel. The DC power supply at CON6 can be provided through an adaptor, or through a battery.

After proper assembly, the switchboard is ready to use. If you feel that frequency of LED switching is too low, you can increase it by using preset VR1.

To test the circuit for proper functioning, check correct power supply at TP1 with respect to TP0. The frequency of the astable multivibrator can be observed at TP2. You can also check the status of switch S2 at TP3. ●

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